

TSELINA, Russian Federation

WMO: 347470

Lat: 46.550N

Lon: 41.050E

Elev: 367

StdP: 14.50

Time Zone: 3.00 (E03)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.3	3.8	-5.9	4.1	-0.2	-1.2	5.2	4.8	27.1	15.2	19.4	30.6	3.0	90	0.445

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	91.8	70.1	88.5	69.1	85.5	67.8	72.4	86.5	70.9	84.1	69.6	82.0	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.3	105.4	77.5	66.8	99.9	76.1	65.4	95.0	74.8	36.3	86.8	35.0	84.4	33.9	82.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.8	14.0	11.6	-7.0	96.0	5.1	3.5	-10.6	98.5	-13.6	100.6	-16.5	102.6	-20.2	105.1		
			-7.4	75.7	5.1	2.7	-11.0	77.7	-14.0	79.3	-16.9	80.8	-20.5	82.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	26.2	25.5	36.5	51.2	61.2	69.1	73.2	71.2	60.8	49.9	36.6	28.5	
	DBStd	19.11	10.45	10.83	9.02	7.30	6.55	6.12	5.21	5.21	7.50	8.79	10.06	9.31	
	HDD50	3122	739	686	428	72	6	0	0	0	9	107	408	667	
	HDD65	6425	1204	1105	885	417	153	28	2	9	168	472	851	1131	
	CDD50	2864	0	0	8	107	352	574	718	658	333	105	8	1	
	CDD65	691	0	0	0	2	35	152	255	202	41	4	0	0	
	CDH74	6594	0	0	1	40	342	1340	2490	2000	344	37	0	0	
	CDH80	2358	0	0	0	8	65	470	1016	719	77	3	0	0	
Wind	WSAvg	4.7	5.5	5.6	6.3	5.7	4.3	3.6	3.1	3.3	3.5	4.5	5.1	5.4	
Precipitation	PrecAvg	21.5	1.6	1.4	1.3	1.8	2.0	2.1	2.3	2.0	1.5	1.3	1.8	2.4	
	PrecMax	28.6	3.9	3.8	2.8	5.4	4.2	4.6	7.5	5.4	6.7	3.1	4.0	6.1	
	PrecMin	13.1	0.2	0.0	0.1	0.2	0.5	0.0	0.7	0.2	0.0	0.1	0.3	0.2	
	PrecStd	4.1	0.9	1.0	0.7	1.3	1.1	1.2	1.5	1.8	1.3	0.8	1.1	1.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	57.2	66.1	78.1	85.4	92.8	97.9	96.0	86.5	78.5	62.4	54.0	
		MCWB	45.0	47.4	52.9	59.2	65.7	71.8	71.3	71.7	68.3	62.9	56.8	50.7	
	2%	DB	43.8	46.6	59.8	72.8	81.4	88.8	93.4	90.2	82.5	72.6	56.6	47.2	
		MCWB	41.9	41.9	48.7	56.9	64.4	69.5	71.1	69.5	66.4	59.7	52.6	44.7	
	5%	DB	40.8	41.2	53.9	68.1	77.6	85.5	89.3	87.1	78.3	67.3	52.3	42.5	
		MCWB	38.9	38.2	45.6	54.5	62.3	68.4	70.1	67.9	63.8	57.3	49.6	40.4	
	10%	DB	37.9	37.4	48.9	64.5	73.8	82.1	85.6	83.9	74.2	62.7	49.0	39.7	
		MCWB	36.2	35.6	43.1	52.5	61.2	67.9	68.4	66.5	62.4	54.8	47.0	38.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.3	47.6	53.2	61.8	68.5	74.5	74.8	73.3	70.7	64.4	56.8	51.4
			MCDB	47.7	56.7	65.7	74.0	80.4	87.8	89.3	89.0	83.1	76.4	61.5	54.3
2%		WB	42.2	42.2	49.2	58.7	66.2	72.0	72.7	71.2	67.8	60.8	53.2	44.8	
		MCDB	43.7	46.3	57.9	69.6	76.8	84.0	87.3	85.6	77.6	70.2	56.3	47.0	
5%		WB	39.1	38.4	46.3	56.2	64.2	70.3	71.1	69.7	65.5	58.3	50.1	40.9	
		MCDB	40.3	40.9	53.3	65.7	74.1	81.6	85.1	83.0	75.2	65.6	51.7	42.3	
10%		WB	36.2	35.8	43.4	53.9	62.3	68.7	69.8	68.2	63.7	56.0	46.9	38.1	
		MCDB	37.4	37.3	48.4	62.4	71.7	79.1	82.9	80.3	72.2	61.8	49.0	39.7	
Mean Daily Temperature Range	5% DB	MDBR	9.7	11.8	13.2	18.1	18.7	18.2	20.2	21.5	19.4	17.0	10.4	9.1	
		MCDBR	10.5	15.0	22.2	25.4	24.3	23.6	25.8	25.9	24.4	23.1	15.1	11.1	
		MCWBR	8.5	10.9	12.9	12.9	11.1	9.4	9.6	10.7	11.7	12.3	10.8	9.2	
	5% WB	MCDBR	10.1	14.5	21.4	22.7	20.6	20.1	22.0	22.1	21.3	20.7	14.0	10.8	
		MCWBR	8.7	10.7	13.0	12.9	10.7	9.3	9.7	10.6	11.5	12.0	10.7	9.3	
Clear-Sky Solar Irradiance	taub	0.341	0.375	0.413	0.453	0.419	0.425	0.450	0.448	0.418	0.386	0.354	0.346		
	taud	2.229	2.141	2.124	2.064	2.250	2.265	2.203	2.202	2.254	2.298	2.318	2.242		
	Ebn at Noon	228	244	253	254	269	268	259	253	250	239	224	211		
	Edh at Noon	27	36	42	49	42	42	44	42	37	30	24	24		
All-Sky Solar Radiation	RadAvg	338	608	935	1473	1887	2031	2017	1818	1327	790	397	265		
	RadStd	71	103	85	143	198	146	100	131	130	102	62	31		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+229	+151

Nomenclature: See separate page